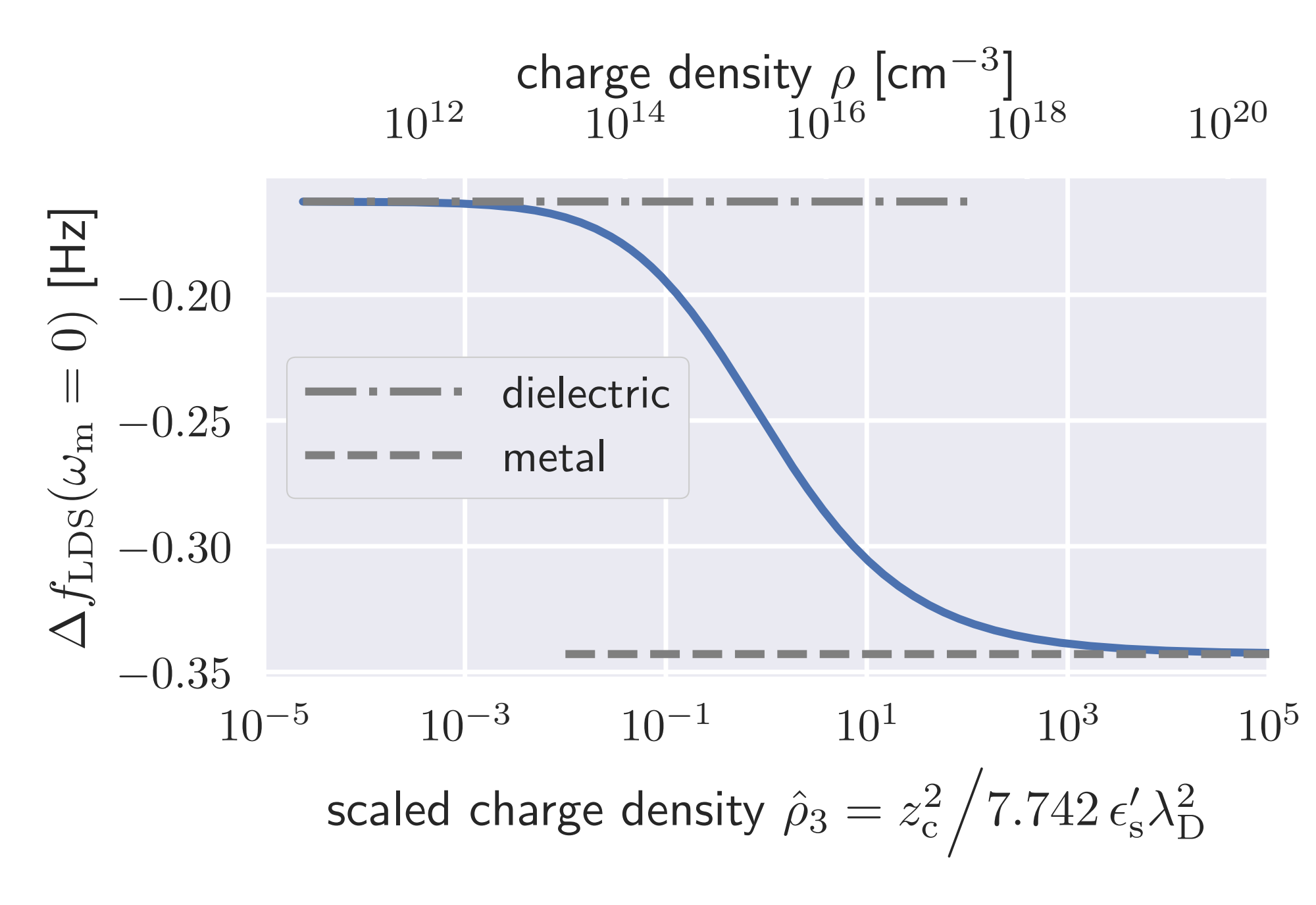

charge density ρ [cm^{-3}]
10^{12}
10^{14}
10^{16}
10^{18}
10^{20}
$\Delta f_{\mathrm{LDS}}(\omega_{\mathrm{m}} = 0)$ [Hz]
-0.20
-0.25
-0.30
-0.35
dielectric
metal
10^{-5}
10^{-3}
10^{-1}
10^{1}
10^{3}
10^{5}
scaled charge density $\hat{\rho}_3 = z_{\mathrm{c}}^2 \Big/ 7.742\,\epsilon'_{\mathrm{s}}\lambda_{\mathrm{D}}^2$